Practical No. 8

1. Create a login form with username and password fields, and validate non-empty input.

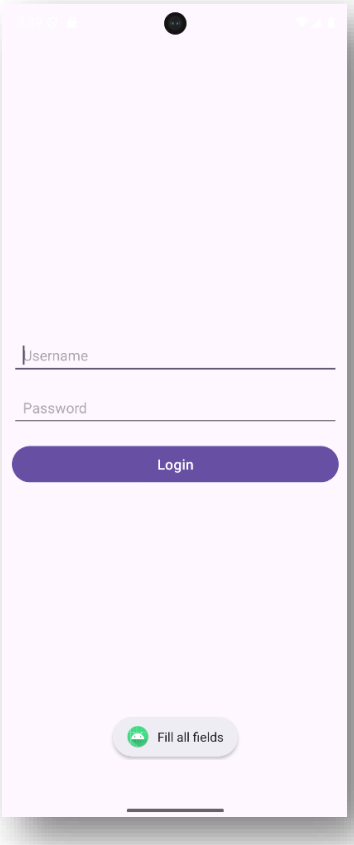
MainActivity.java :

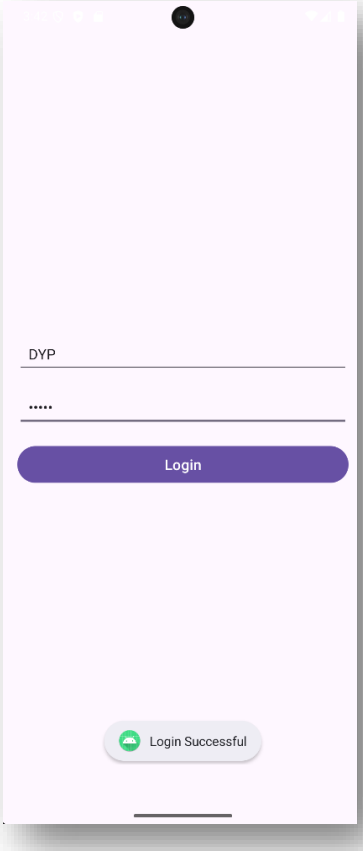
```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   protected void onCreate(Bundle b) {
      super.onCreate(b);
      setContentView(R.layout.activity_main);

      EditText u = findViewById(R.id.etUser);
      EditText p = findViewById(R.id.etPass);
      Button btn = findViewById(R.id.btnLogin);

      btn.setOnClickListener(v -> {
         if(u.getText().toString().isEmpty() || p.getText().toString().isEmpty())
            Toast.makeText(this,"Fill all fields",Toast.LENGTH_SHORT).show();
         else if(u.getText().toString().equals("DYP") && p.getText().toString().equals("login"))
            Toast.makeText(this,"Login Successful",Toast.LENGTH_SHORT).show();
         else
            Toast.makeText(this,"Invalid Credentials",Toast.LENGTH_SHORT).show();
      });
   }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp">

   <EditText
      android:id="@+id/etUser"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
```

```
        android:hint="Username"
        android:layout_marginBottom="12dp"
        android:padding="12dp"
        android:textSize="16sp"/>

    <EditText
        android:id="@+id/etPass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginBottom="16dp"
        android:padding="12dp"
        android:textSize="16sp"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="16sp"/>
</LinearLayout>
```





2. Add gender selection using radio buttons and display the selected gender on login.

MainActivity.java :

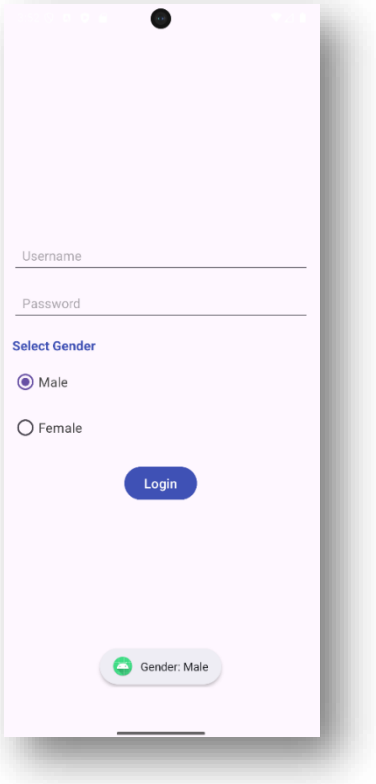
```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   protected void onCreate(Bundle b) {
      super.onCreate(b);
      setContentView(R.layout.activity_main);

      EditText u = findViewById(R.id.etUser);
      EditText p = findViewById(R.id.etPass);
      Button btn = findViewById(R.id.btnLogin);

      RadioButton m = findViewById(R.id.rbMale);
      RadioButton f = findViewById(R.id.rbFemale);

      btn.setOnClickListener(v -> {
         if(u.getText().toString().isEmpty() || p.getText().toString().isEmpty())
            Toast.makeText(this,"Fill all fields",Toast.LENGTH_SHORT).show();
         else if(u.getText().toString().equals("DYP") && p.getText().toString().equals("login"))
            Toast.makeText(this,"Login Successful",Toast.LENGTH_SHORT).show();
         else
            Toast.makeText(this,"Invalid Credentials",Toast.LENGTH_SHORT).show();
      });
      btn.setOnClickListener(v -> {
            if(m.isChecked())
               Toast.makeText(this,"Gender: Male",Toast.LENGTH_SHORT).show();
            if(f.isChecked())
               Toast.makeText(this,"Gender: Female",Toast.LENGTH_SHORT).show();
         });
         }
      }
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
```

```
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <EditText
        android:id="@+id/etUser"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:layout_marginBottom="12dp"
        android:padding="12dp"
        android:textSize="16sp"/>

    <EditText
        android:id="@+id/etPass"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginBottom="18dp"
        android:padding="12dp"
        android:textSize="16sp"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Gender"
        android:textSize="16sp"
        android:textStyle="bold"
        android:textColor="#3F51B5"
        android:layout_marginBottom="10dp"/>

    <RadioGroup
        android:id="@+id/rg"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="20dp">

        <RadioButton
            android:id="@+id/rbMale"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Male"
            android:textSize="16sp"
            android:textColor="#444444"
            android:buttonTint="#3F51B5"
            android:layout_marginBottom="8dp"/>
```

```
    <RadioButton
      android:id="@+id/rbFemale"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Female"
      android:textSize="16sp"
      android:textColor="#444444"
      android:buttonTint="#3F51B5"/>
  </RadioGroup>

  <Button
    android:id="@+id/btnLogin"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login"
    android:textSize="16sp"
    android:textColor="#FFFFFF"
    android:backgroundTint="#3F51B5"/>
</LinearLayout>
```



3 .Implement a "Remember Me" checkbox and show a toast message based on its state.

MainActivity.java :

```
package com.techtitans.madpracticals;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  protected void onCreate(Bundle b) {
    super.onCreate(b);
    setContentView(R.layout.activity_main);

    EditText user = findViewById(R.id.etUser);
    EditText pass = findViewById(R.id.etPass);
    Button btn = findViewById(R.id.btnLogin);

    RadioButton male = findViewById(R.id.rbMale);
    RadioButton female = findViewById(R.id.rbFemale);
    CheckBox check = findViewById(R.id.chk);
```

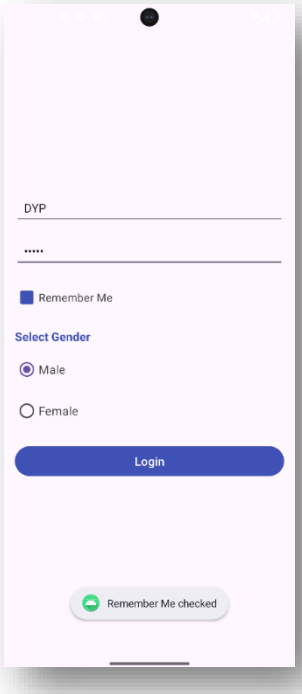
```
        btn.setOnClickListener(v -> {

            if(user.getText().toString().isEmpty() || pass.getText().toString().isEmpty()) {
                Toast.makeText(this,"Fill all fields",Toast.LENGTH_SHORT).show();
                return;
            }

            if(user.getText().toString().equals("DYP") && pass.getText().toString().equals("login"))
                Toast.makeText(this,"Login Successful",Toast.LENGTH_SHORT).show();
            else {
                Toast.makeText(this,"Invalid Credentials",Toast.LENGTH_SHORT).show();
                return;
            }

            if(male.isChecked())
                Toast.makeText(this,"Gender: Male",Toast.LENGTH_SHORT).show();
            else if(female.isChecked())
                Toast.makeText(this,"Gender: Female",Toast.LENGTH_SHORT).show();

            if(check.isChecked())
                Toast.makeText(this,"Remember Me checked",Toast.LENGTH_SHORT).show();
            else {
                Toast.makeText(this,"Remember Me Unchecked",Toast.LENGTH_SHORT).show();

            }
        });
    }
}
```

activity_main.xml :

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <EditText
        android:id="@+id/etUser"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:layout_marginBottom="12dp"
        android:padding="12dp"
        android:textSize="16sp"/>
```

```
<EditText
    android:id="@+id/etPass"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:layout_marginBottom="16dp"
    android:padding="12dp"
    android:textSize="16sp"/>

<CheckBox
    android:id="@+id/chk"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Remember Me"
    android:textSize="15sp"
    android:textColor="#444444"
    android:buttonTint="#3F51B5"
    android:layout_marginBottom="18dp"/>

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Select Gender"
    android:textSize="16sp"
    android:textStyle="bold"
    android:textColor="#3F51B5"
    android:layout_marginBottom="10dp"/>

<RadioGroup
    android:id="@+id/rg"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="20dp">

    <RadioButton
        android:id="@+id/rbMale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Male"
        android:textSize="16sp"
        android:textColor="#444444"
        android:buttonTint="#3F51B5"
        android:layout_marginBottom="8dp"/>

    <RadioButton
```

```
        android:id="@+id/rbFemale"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Female"
        android:textSize="16sp"
        android:textColor="#444444"
        android:buttonTint="#3F51B5"/>
    </RadioGroup>

    <Button
      android:id="@+id/btnLogin"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Login"
      android:textSize="16sp"
      android:textColor="#FFFFFF"
      android:backgroundTint="#3F51B5"/>
</LinearLayout>
```



4. Design a form that disables the login button until all fields are filled

MainActivity.java :

```
package com.techtitans.madpracticals;
import android.os.Bundle;
import android.text.*;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {

    protected void onCreate(Bundle b) {
      super.onCreate(b);
      setContentView(R.layout.activity_main);

      EditText user = findViewById(R.id.etUser);
      EditText pass = findViewById(R.id.etPass);
      Button btn = findViewById(R.id.btnLogin);
      RadioButton male = findViewById(R.id.rbMale);
      RadioButton female = findViewById(R.id.rbFemale);
      CheckBox check = findViewById(R.id.chk);

      btn.setEnabled(false);
      btn.setBackgroundTintList(
            android.content.res.ColorStateList.valueOf(
                  android.graphics.Color.parseColor("#BDBDBD")));
      user.addTextChangedListener(new TextWatcher() {
```

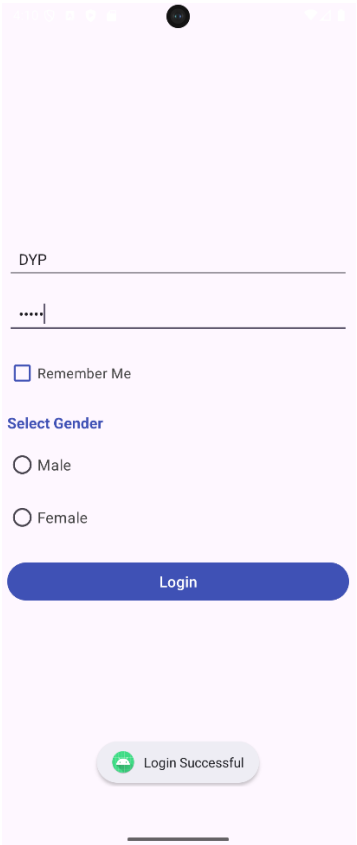
```
    @Override
    public void afterTextChanged(Editable s) {
        if (!user.getText().toString().isEmpty() &&
                !pass.getText().toString().isEmpty()) {

            btn.setEnabled(true);
            btn.setBackgroundTintList(
                    android.content.res.ColorStateList.valueOf(
                            android.graphics.Color.parseColor("#3F51B5")));
        } else {
            btn.setEnabled(false);
            btn.setBackgroundTintList(
                    android.content.res.ColorStateList.valueOf(
                            android.graphics.Color.parseColor("#BDBDBD")));
    }           }
    @Override
    public void beforeTextChanged(CharSequence s, int start, int count, int after) { }
    @Override
    public void onTextChanged(CharSequence s, int start, int before, int count) { }
});

pass.addTextChangedListener(new TextWatcher() {
    @Override
    public void afterTextChanged(Editable s) {
        if (!user.getText().toString().isEmpty() &&
                !pass.getText().toString().isEmpty()) {

            btn.setEnabled(true);
            btn.setBackgroundTintList(
                    android.content.res.ColorStateList.valueOf(
                            android.graphics.Color.parseColor("#3F51B5")));

        } else {
            btn.setEnabled(false);
            btn.setBackgroundTintList(
                    android.content.res.ColorStateList.valueOf(
                            android.graphics.Color.parseColor("#BDBDBD")));
        }           }
    @Override
    public void beforeTextChanged(CharSequence s, int start, int count, int after) { }

    @Override
    public void onTextChanged(CharSequence s, int start, int before, int count) { }
});


btn.setOnClickListener(v -> {
```

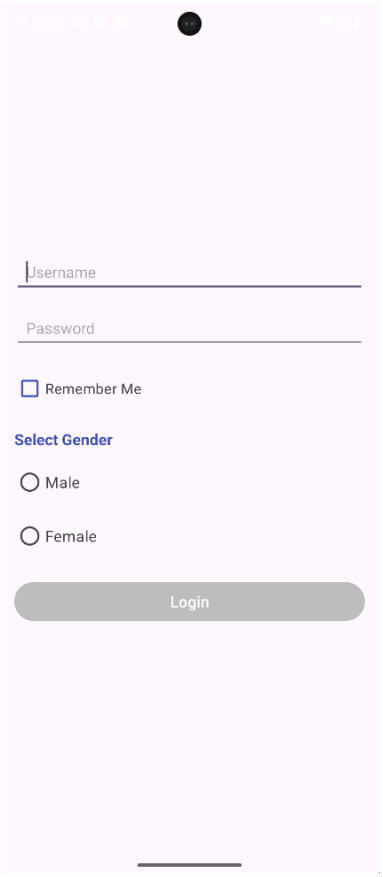
```
        if(user.getText().toString().equals("DYP") &&
            pass.getText().toString().equals("login"))
          Toast.makeText(this,"Login Successful",Toast.LENGTH_SHORT).show();
        else {
          Toast.makeText(this,"Invalid Credentials",Toast.LENGTH_SHORT).show();
          return;
        }

        if(male.isChecked())
          Toast.makeText(this,"Gender: Male",Toast.LENGTH_SHORT).show();
        else if(female.isChecked())
          Toast.makeText(this,"Gender: Female",Toast.LENGTH_SHORT).show();

        if(check.isChecked())
          Toast.makeText(this,"Remember Me checked",Toast.LENGTH_SHORT).show();
        else
          Toast.makeText(this,"Remember Me Unchecked",Toast.LENGTH_SHORT).show();
      });
    }
}
```

activity_main.xml :

Chnageable part only :-

```
<Button
  android:id="@+id/btnLogin"
  android:layout_width="match_parent"
  android:layout_height="wrap_content"
  android:text="Login"
  android:textSize="16sp"
  android:textColor="#FFFFFF"
  android:backgroundTint="#BDBDBD"
  android:enabled="false"/>
```





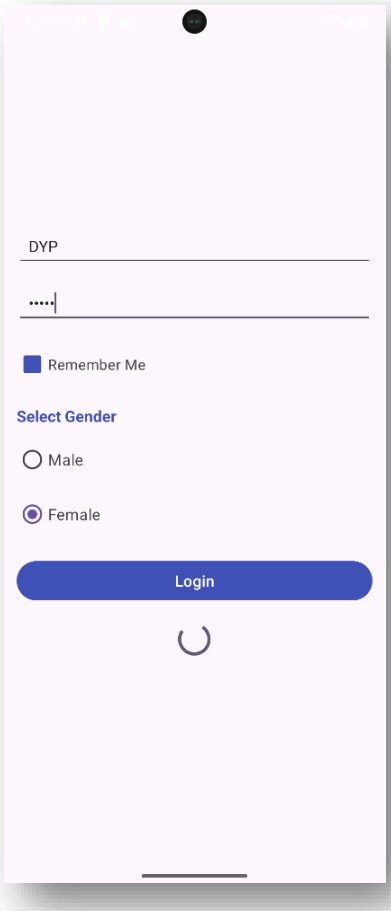
5 .Modify the form to include a progress bar that appears during login processing.

MainActivity.java :

```
package com.techtitans.madpracticals;

import android.os.*;
import android.text.*;
import android.view.View;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  protected void onCreate(Bundle b) {
    super.onCreate(b);
    setContentView(R.layout.activity_main);

    EditText user = findViewById(R.id.etUser);
    EditText pass = findViewById(R.id.etPass);
    Button btn = findViewById(R.id.btnLogin);

    RadioButton male = findViewById(R.id.rbMale);
    RadioButton female = findViewById(R.id.rbFemale);
    CheckBox check = findViewById(R.id.chk);
    ProgressBar pr = findViewById(R.id.pr);

    btn.setEnabled(false);
    pr.setVisibility(View.INVISIBLE);

    user.addTextChangedListener(new TextWatcher() {
      public void afterTextChanged(Editable s) {
        if(!user.getText().toString().isEmpty() &&
          !pass.getText().toString().isEmpty()) {
          btn.setEnabled(true);
          btn.setBackgroundTintList(
              android.content.res.ColorStateList.valueOf(
                  android.graphics.Color.parseColor("#3F51B5")));
        } else {
          btn.setEnabled(false);
          btn.setBackgroundTintList(
              android.content.res.ColorStateList.valueOf(
                  android.graphics.Color.parseColor("#BDBDBD")));
        }
      }
      public void beforeTextChanged(CharSequence a,int b,int c,int d){}
      public void onTextChanged(CharSequence a,int b,int c,int d){}
    });

    pass.addTextChangedListener(new TextWatcher() {
```

```
        public void afterTextChanged(Editable s) {
            if(!user.getText().toString().isEmpty() &&
              !pass.getText().toString().isEmpty()) {
                btn.setEnabled(true);
                btn.setBackgroundTintList(
                        android.content.res.ColorStateList.valueOf(
                                android.graphics.Color.parseColor("#3F51B5")));
            } else {
                btn.setEnabled(false);
                btn.setBackgroundTintList(
                        android.content.res.ColorStateList.valueOf(
                                android.graphics.Color.parseColor("#BDBDBD")));
            }
        }
        public void beforeTextChanged(CharSequence a,int b,int c,int d){}
        public void onTextChanged(CharSequence a,int b,int c,int d){}
    });

    btn.setOnClickListener(v -> {

        if(!user.getText().toString().equals("DYP") ||
          !pass.getText().toString().equals("login")) {
            Toast.makeText(this,"Invalid Credentials",Toast.LENGTH_SHORT).show();
            return;
        }

        btn.setEnabled(false);
        pr.setVisibility(View.VISIBLE);

        new Handler(Looper.getMainLooper()).postDelayed(() -> {

            pr.setVisibility(View.INVISIBLE);
            btn.setEnabled(true);

            Toast.makeText(this,"Login Successful",Toast.LENGTH_SHORT).show();

            if(male.isChecked())
                Toast.makeText(this,"Gender: Male",Toast.LENGTH_SHORT).show();
            else if(female.isChecked())
                Toast.makeText(this,"Gender: Female",Toast.LENGTH_SHORT).show();

            if(check.isChecked())
                Toast.makeText(this,"Remember Me checked",Toast.LENGTH_SHORT).show();

        }, 4000);
    });
  }
}
```

activity_main.xml :

changeable code only :

```
<Button
   android:id="@+id/btnLogin"
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:text="Login"
   android:textSize="16sp"
   android:textColor="#FFFFFF"
   android:backgroundTint="#BDBDBD"
   android:enabled="false"/>

<ProgressBar
   android:id="@+id/pr"
   android:layout_marginTop="16dp"
   android:visibility="invisible"
   android:layout_height="40dp"
   android:layout_width="40dp"/>
```